

The empirical recurrence for OEIS A235769: recovering a conjecture that is not written down

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Abstract

OEIS A235769 records “Empirical recurrence of order 88”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1202$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 88, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 88 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235769 is “Number of $(n+1) \times (4+1)$ 0..1 arrays with the sum of each 2×2 subblock two extreme terms minus its two median terms lexicographically nondecreasing columnwise and nonincreasing rowwise.”. It has offset 1 and begins

288, 1948, 12417, 70752, 377890, 1874171, 8783904, 38884279, ...

The entry states:

Empirical recurrence of order 88 (see link above).

As of the “Last modified” line on the live entry (Jun 26 2022 03:42:42, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1522$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 1202$ of the 1522, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1202$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 3044$ exact terms of the model returns order 88 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	5	c_{23}	−728900854416	c_{45}	−356379857585848	c_{67}	263913768471552
c_2	52	c_{24}	1079718509772	c_{46}	−6031960820661076	c_{68}	77604672589824
c_3	−298	c_{25}	2470629148596	c_{47}	1038165882300300	c_{69}	−111299280771072
c_4	−1264	c_{26}	−4191355646856	c_{48}	7088495452832051	c_{70}	−22008088657920
c_5	8628	c_{27}	−7379111962656	c_{49}	−1849484745063727	c_{71}	40673725157376
c_6	18692	c_{28}	14270774723424	c_{50}	−7529858691152816	c_{72}	4666942939136
c_7	−161668	c_{29}	19456365277584	c_{51}	2598856226912666	c_{73}	−12743058964480
c_8	−179318	c_{30}	−42994953949128	c_{52}	7219777723394168	c_{74}	−540559278080
c_9	2203502	c_{31}	−45269716891272	c_{53}	−3080614330987668	c_{75}	3374276182016
c_{10}	1005308	c_{32}	115370168303769	c_{54}	−6235028967960384	c_{76}	−71546306560
c_{11}	−23279496	c_{33}	92628557591331	c_{55}	3164517301106424	c_{77}	−740721229824
c_{12}	−57952	c_{34}	−277084054410972	c_{56}	4835693926838832	c_{78}	57509937152
c_{13}	198361064	c_{35}	−165288490610958	c_{57}	−2852693906453952	c_{79}	131213164544
c_{14}	−67567268	c_{36}	597852129613856	c_{58}	−3354954617158656	c_{80}	−16804020224
c_{15}	−1400448580	c_{37}	252667033123484	c_{59}	2270126153794176	c_{81}	−18025480192
c_{16}	865749269	c_{38}	−1162145616755948	c_{60}	2071427942453760	c_{82}	3125805056
c_{17}	8351539623	c_{39}	−317663140406692	c_{61}	−1598204963397888	c_{83}	1802108928
c_{18}	−7150724328	c_{40}	2039396295845018	c_{62}	−1130295973644288	c_{84}	−386400256
c_{19}	−42669582690	c_{41}	291672035479806	c_{63}	995076634037760	c_{85}	−116654080
c_{20}	45721279128	c_{42}	−3235357683376180	c_{64}	539892162823680	c_{86}	29360128
c_{21}	188771839620	c_{43}	−88707692539096	c_{65}	−546668778152448	c_{87}	3670016
c_{22}	−240951098136	c_{44}	4643781760622272	c_{66}	−222658634588160	c_{88}	−1048576

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 88, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $88 + 4$ terms removed returns order 88 as well, so $\deg q_{\min} = 88 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $88 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 88 that this sequence satisfies — and that family turns out to have exactly one member.

References

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